

CS1231S Condensed Notes · Revision Addendum

This addendum compiles all content added or refined after the initial commit (062d28e), including lecture definitions, equivalence relation notes, well-defined operations on quotient sets, selected tutorial proofs (Part VI), and updated index terms. The material is also integrated into the current condensed notes; keep this companion alongside an initial printed copy. Tutorial solutions are adapted from the supplied answers, with explicit domains and justifications.

Part I · Logic & Arguments (§1, §3, §4)

DEFINITION Negation (2.1.2) · Conjunction (2.1.3) · Disjunction (2.1.4) · §1

For statement variables p and q : the **negation** of p is “not p ” or “it is not the case that p ”, denoted $\sim p$; the **conjunction** of p and q is “ p and q ”, denoted $p \wedge q$; the **disjunction** of p and q is “ p or q ”, denoted $p \vee q$ (inclusive: true when either or both are true).

METHOD Clarification on §1 & §3 Wording

- **Tautological / Contradictory statements (§1)**: A statement whose statement form is a tautology (contradiction) is called a **tautological (contradictory)** statement.
- **Conditional terminology (§3)**: In $p \rightarrow q$ (“if p then q ”, “ p implies q ”), p is called the **hypothesis** (or **antecedent**) and q is called the **conclusion** (or **consequent**).

DEFINITION Argument, validity (2.3.1) · Critical row · Syllogism · Sound, unsound (2.3.2) · §4

An **argument** (**argument form**) is a sequence of statements (statement forms); all but the last are **premises** (assumptions, hypotheses), the last (after $\therefore$) is the **conclusion**.

An argument form is **valid** iff, whatever statements are substituted for its variables, if the premises are all true then the conclusion is true.

A **critical row** is a truth-table row in which all premises are T: the form is valid iff the conclusion is T in every critical row.

A **syllogism** is an argument form with two premises and a conclusion (e.g. modus ponens).

An argument is **sound** iff it is valid **and** all its premises are true; otherwise it is **unsound**.

Part II · Predicates, Quantifiers & Proofs (§5, §8, §10)

DEFINITION Domain of discourse & Truth set (§5 update)

The **domain** of a variable is also termed the **domain of discourse**, **universe of discourse**, **universal set**, or **universe**. The **truth set** of $P(x)$, $x \in D$, is the set of all elements of D that make $P(x)$ true: $\{x \in D : P(x)\}$, also denoted $\{x \in D \mid P(x)\}$.

DEFINITION Kinds of mathematical statements (Lecture 1) · §5

- A **universal statement** says a property holds for **all** elements of a set ($\forall$; “all”, “every”, “any”).
- A **conditional statement** says that if one thing is true then another must be ($\rightarrow$; “if ... then”).
- An **existential statement** says there is **at least one** thing with the property ($\exists$; “there exists”, “some”).
- A **universal conditional statement** is both universal and conditional: $\forall x (P(x) \rightarrow Q(x))$ (e.g. “for all animals a , if a is a dog then a is a mammal”).
- A **universal existential statement** has the form $\forall x \exists y \dots$ (e.g. “every real number has an additive inverse”).
- An **existential universal statement** has the form $\exists x \forall y \dots$ (e.g. “there is a positive integer that is $\leq$ every positive integer”).

DEFINITION Valid argument form with quantifiers (3.4.1) · §8

An argument form is **valid** iff, no matter what particular predicates are substituted for the predicate symbols in its premises, if the resulting premise statements are all true then the conclusion is also true. An argument is **valid** iff its form is valid.

DEFINITION Absolute value (Lecture 1) · Colorful (Lecture 1, CS1231S only) · §10

For $x \in \mathbb{R}$, the **absolute value** of x is $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.

An integer n is **colorful** iff there exists some integer k such that $n = 3k$. (Used in lecture: $-1353 = 3 \cdot (-451)$ and $0 = 3 \cdot 0$ are colorful; 7 is not.)

METHOD Axioms & Primes (§10 updates)

- **Axiom / postulate:** Assumed true without proof (Peano axioms, Euclid's postulates).
- **Equivalent prime definition:** $n \in \mathbb{Z}$ with $n > 1$ is prime iff $\forall r, s \in \mathbb{Z} (r > 1 \wedge s > 1 \rightarrow rs \neq n)$.

Part III · Sets & Operations (§13, §14, §16)**DEFINITION Set-roster notation & Cardinality (§13 update)**

- **Set-roster:** list all elements between braces: $\{1, 2, 3\}$, $\{1, 2, \dots, 100\}$, $\{1, 2, 3, \dots\}$ (the ellipsis “...” reads “and so forth”).
- **Set elements:** $x \in S$ (x is a **member** or **element**); $x \notin S$ means it is not.
- **Cardinality** $|S|$: the size of S , the number of elements of S .

DEFINITION Interval notation (Lecture 5) · §13

For real numbers $a \leq b$:

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}, \quad [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\},$$

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}, \quad [a, b) = \{x \in \mathbb{R} : a \leq x < b\}.$$

Unbounded intervals:

$$(a, \infty) = \{x \in \mathbb{R} : x > a\}, \quad [a, \infty) = \{x \in \mathbb{R} : x \geq a\},$$

$$(-\infty, b) = \{x \in \mathbb{R} : x < b\}, \quad (-\infty, b] = \{x \in \mathbb{R} : x \leq b\}.$$

Note: The interval (a, b) and the ordered pair (a, b) look identical; context distinguishes them.

DEFINITION Superset · Singleton (§14 update)

- If $A \subseteq B$, we write $B \supseteq A$: “ B contains / includes A ”, “ B is a **superset** of A ”.
- $A \subsetneq B$ denotes **proper** (strict) inclusion.
- The **empty set** $\emptyset = \{\}$ has no elements; a set with exactly one element is a **singleton**.

DEFINITION Universal set · Relative complement (§16 update)

- A **universal set** (universe of discourse) U contains every object under discussion (e.g. $\mathbb{R}$ when all sets are sets of reals).
- $B \setminus A = \{x \in U : x \in B \wedge x \notin A\}$ is also called the **relative complement** of A in B .

Part IV · Relations, Digraphs & Equivalence (§19, §23)

DEFINITION Relation notation · Directed graph · n -ary relations (§19 update)

- xRy means “ x is R -related to y ”; $x\cancel{R}y$ means $(x, y) \notin R$.
- A **(binary) relation on A** is a subset of $A \times A = A^2$; an n -ary relation on A is a subset of A^n .
- The **directed graph** (digraph) of R on A draws each element of A once as a vertex, with a directed arrow $x \rightarrow y$ whenever xRy and a self-loop at x when xRx .
- An **n -ary relation** on $A_1 \times \cdots \times A_n$ is a subset of the Cartesian product (binary, ternary, quaternary for $n = 2, 3, 4$).

DEFINITION Procedural class definition · Congruence modulo n (Lecture 6.3.4) · §23

- Procedural equivalence class: $\forall x \in A (x \in [a]_{\sim} \leftrightarrow a \sim x)$.
- Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$. a is **congruent to b modulo n** , written $a \equiv b \pmod{n}$, iff $a - b = nk$ for some $k \in \mathbb{Z}$ (i.e. $n \mid (a - b)$).
- Classes: $[x] = \{x + nk : k \in \mathbb{Z}\}$, and $\mathbb{Z}/\sim_n = \{[0], [1], \dots, [n-1]\}$.

DEFINITION Representative · Well-defined operation on $A/\sim$ · §23

Any $b \in [a]$ is a **representative** of $[a]$, since then $[b] = [a]$: mod 5, $[2] = [7] = [12]$.

Let $\sim$ be an equivalence relation on A and $*$ a binary operation on A (so $a * b \in A$ for all $a, b \in A$).

The induced rule $[a] * [b] = [a * b]$ is **well defined** iff the answer depends only on the classes, not on the representatives chosen:

$$a \sim a' \wedge b \sim b' \Rightarrow a * b \sim a' * b'.$$

PROOF $+$ and $\cdot$ are well defined on the classes of congruence mod n · §23

1. Fix $n \in \mathbb{Z}^+$ and let $a, a', b, b' \in \mathbb{Z}$ with $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$: $a - a' = nk$ and $b - b' = nm$ for some $k, m \in \mathbb{Z}$ (definition of congruence).
2. (Addition) $(a + b) - (a' + b') = n(k + m)$ with $k + m \in \mathbb{Z}$ (closure), so $a + b \equiv a' + b' \pmod{n}$.
3. (Multiplication) $ab - a'b' = (a - a')b + a'(b - b') = n(kb + a'm)$ with $kb + a'm \in \mathbb{Z}$ (closure), so $ab \equiv a'b' \pmod{n}$. ■

E.g. mod 5: $[2] \cdot [4] = [8] = [3]$ and $[7] \cdot [9] = [63] = [3]$.

TRAP Check reflexive, symmetric, transitive first · then check well-definedness · §23

- $a \sim b \leftrightarrow ab \geq 0$ is **not** transitive on $\mathbb{R}$ ($1 \sim 0$ and $0 \sim -1$, but $1 \not\sim -1$), so it does not give a partition into equivalence classes. On $\mathbb{R} \setminus \{0\}$ it means “same sign”: $(\mathbb{R} \setminus \{0\})/\sim = \{[1], [-1]\} = \{(0, \infty), (-\infty, 0)\}$.
- $a \sim b \leftrightarrow a - b = 2k$ for some $k \in \mathbb{Z}$, on $\mathbb{R}$: $[a] = \{a + 2k : k \in \mathbb{Z}\}$ and $\mathbb{R}/\sim = \{[r] : 0 \leq r < 2\}$ (infinitely many classes). $+$ is well defined, but $\cdot$ is **not**: $[0.5] = [2.5]$, yet $[0.5] \cdot [0.5] = [0.25] \neq [1.25] = [2.5] \cdot [0.5]$, since $1.25 - 0.25 = 1$ is not of the form $2k$.

Part V · Partial Orders & Well-Ordered Sets (§24)

DEFINITION Maximal and minimal alternate characterizations (§24 update)

- c **maximal** $\Leftrightarrow \forall x \in A (c \preceq x \rightarrow c = x)$. Equivalently, every $x \in A$ has $x \preceq c$ or is not comparable with c .
- c **minimal** $\Leftrightarrow \forall x \in A (x \preceq c \rightarrow c = x)$. Equivalently, every $x \in A$ has $c \preceq x$ or is not comparable with c .

DEFINITION Well-ordered set (Lecture 6.4.7) · §24

Let $\preccurlyeq$ be a total order on a set A . A is **well-ordered** iff every non-empty subset of A contains a smallest element. Symbolically,

$$\forall S \in \mathcal{P}(A) (S \neq \emptyset \rightarrow \exists x \in S \forall y \in S (x \preccurlyeq y)).$$

$(\mathbb{N}, \leq)$ is well-ordered. $(\mathbb{Z}, \leq)$ is not: $\mathbb{Z}$ itself (or $\mathbb{Z}^-$) is a non-empty subset with no smallest element.

Theorem: Every total order on a finite set is well-ordered (a non-empty subset has a minimal element, which in a total order is smallest).

Part VI · Tutorial Worked Examples (Tutorials 1–5)

Model solutions adapted from Tutorials 1–5, in the numbered-line, justified style of the main notes. Each illustrates a standard proof pattern. Results already covered by the initial notes ($A \subseteq B \leftrightarrow A \cup B = B$, associativity of $\circ$, $\mathbb{Q}$ closed under $+$, R symmetric $\leftrightarrow R = R^{-1}$, AY23 Q12) are omitted here. The tutorial locator in §29 of the current condensed notes gives the integrated locations.

Proof techniques (Tutorials 1–2)

WORKED EXAMPLE T1 Q11 · Biconditional: n^2 is odd $\leftrightarrow n$ is odd

For $n \in \mathbb{Z}$, Smart writes: “**Proof (by contradiction)**. Suppose n is even, so $n = 2k$, ... so n^2 is even. So if n is even then n^2 is even, i.e. if n^2 is odd then n is odd. Therefore n^2 is odd iff n is odd.” Comment, then give a correct proof.

Let $n \in \mathbb{Z}$. **Faults:** (1) only one direction (n^2 odd $\rightarrow n$ odd) is proved; (2) the method is **contraposition**, not contradiction; (3) justifications are missing (definition of even, closure).

1. ($\Rightarrow$) Prove the contrapositive of “if n^2 is odd, then n is odd”.
 - 1.1. Suppose n is even. Then $\exists k \in \mathbb{Z}$ s.t. $n = 2k$ (definition of even).
 - 1.2. $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ (basic algebra), with $2k^2 \in \mathbb{Z}$ (closure of integers under $\times$).
 - 1.3. So n^2 is even (definition of even). This proves: if n^2 is odd, then n is odd (contraposition and the even/odd dichotomy).
2. ($\Leftarrow$) Suppose n is odd: $n = 2k + 1$ for some $k \in \mathbb{Z}$ (definition of odd). Then $n^2 = 2(2k^2 + 2k) + 1$, with $2k^2 + 2k \in \mathbb{Z}$ (basic algebra, closure), so n^2 is odd (definition of odd).
3. Therefore n^2 is odd iff n is odd (lines 1 and 2). ■

WORKED EXAMPLE T2 Q8 · Cases inside a conditional

Prove $\forall x \in \mathbb{R} ((x^2 > x) \rightarrow ((x < 0) \vee (x > 1)))$. Must the case $x^2 \leq x$ be examined?

No: $p \rightarrow q$ is true whenever p is false, so only $x^2 > x$ needs examining.

1. Let r be an arbitrarily chosen real number.
2. Suppose $r^2 > r$.
 - 2.1. Then $r^2 - r > 0$, i.e. $r(r - 1) > 0$ (basic algebra).
 - 2.2. So r and $r - 1$ are both positive or both negative (Appendix A, T25).
 - 2.3. **Case 1:** both positive. Then $r - 1 > 0$, so $r > 1$ (basic algebra).
 - 2.4. **Case 2:** both negative. Then $r < 0$ (case assumption).
 - 2.5. Hence $(r > 1) \vee (r < 0)$; so $(r^2 > r) \rightarrow ((r > 1) \vee (r < 0))$ (division into cases).
3. Therefore $\forall x \in \mathbb{R} ((x^2 > x) \rightarrow ((x < 0) \vee (x > 1)))$ (universal generalization, as r was arbitrary). ■

Do not use diagrams (e.g. graphs) as proofs unless instructed.

PAST PAPER T2 Q11 · Past midterm · A factor is at most $\sqrt{n}$

For $n, a, b \in \mathbb{Z}^+$, prove $n = ab \rightarrow (a \leq \sqrt{n} \vee b \leq \sqrt{n})$. **By contraposition** ($p \rightarrow q \equiv \sim q \rightarrow \sim p$):

1. Contrapositive: if $a > \sqrt{n}$ and $b > \sqrt{n}$, then n is not the product ab (De Morgan’s law).
2. Suppose $a > \sqrt{n}$ and $b > \sqrt{n}$. Then $ab > \sqrt{n} \cdot \sqrt{n} = n$ (Appendix A, T27).
3. Since $ab \neq n$, the contrapositive is true; therefore the original statement is true. ■

By contradiction ($\sim(p \rightarrow q) \equiv p \wedge \sim q$):

1. Suppose not: $n = ab$ and $a > \sqrt{n}$ and $b > \sqrt{n}$ (De Morgan’s law).
2. Then $ab > \sqrt{n} \cdot \sqrt{n} = n$ (Appendix A, T27), contradicting $n = ab$ in line 1. ■

T27: if $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.

TRAP T2 Q4 · Closure claims

$\mathbb{Z}$ is **not** closed under division ($3/2 = 1.5 \notin \mathbb{Z}$). Division is not a binary operation on **all** of $\mathbb{Q} \times \mathbb{Q}$, since $r/0$ is undefined; division of a rational by a **nonzero** rational does give a rational. In proving $\mathbb{Q}$ closed under $+$, the line “ $bd \neq 0$ (zero product property)” is required, and is commonly omitted.

Sets (Tutorial 3)**WORKED EXAMPLE T3 Q4 · Set equality by double inclusion**

$A = \{2n + 1 : n \in \mathbb{Z}\}$, $B = \{2n - 5 : n \in \mathbb{Z}\}$. Is $A = B$? Prove it.

1. ($\subseteq$) Let $a \in A$.
 - 1.1. Use the definition of A to find an integer n such that $a = 2n + 1$.
 - 1.2. Then $a = 2n + 1 = 2(n + 3) - 5$, and $n + 3 \in \mathbb{Z}$ (closure of integers under $+$).
 - 1.3. Therefore $a \in B$ (definition of B).
2. ($\supseteq$) Let $b \in B$; find $n \in \mathbb{Z}$ with $b = 2n - 5$. Then $b = 2(n - 3) + 1$ with $n - 3 \in \mathbb{Z}$ (closure under $-$), so $b \in A$ (definition of A).
3. Therefore $A = B$ (definition of set equality). ■

The change of variables $k = n + 3$ is valid only after showing it ranges over **all** integers: for each $k \in \mathbb{Z}$, take $n = k - 3 \in \mathbb{Z}$. Closure alone does not justify that step; double inclusion makes both directions explicit.

WORKED EXAMPLE T3 Q5 · Element method as a set-builder chain

Prove that for all sets A, B, C : $A \cap (B \setminus C) = (A \cap B) \setminus C$.

$$\begin{aligned}
 A \cap (B \setminus C) &= \{x : x \in A \wedge x \in (B \setminus C)\} \quad (\text{definition of } \cap) \\
 &= \{x : x \in A \wedge (x \in B \wedge x \notin C)\} \quad (\text{definition of } \setminus) \\
 &= \{x : (x \in A \wedge x \in B) \wedge x \notin C\} \quad (\text{associativity of } \wedge) \\
 &= \{x : x \in A \cap B \wedge x \notin C\} \quad (\text{definition of } \cap) \\
 &= (A \cap B) \setminus C \quad (\text{definition of } \setminus). \quad \blacksquare
 \end{aligned}$$

WORKED EXAMPLE T3 Q6 · Symmetric difference by set identities only

(a) $A = \{1, 4, 9, 16\}$, $B = \{2, 4, 6, \dots, 16\}$: find $A \oplus B$. (b) Using Theorem 6.2.2, prove $A \oplus B = (A \cup B) \setminus (A \cap B)$.

(a) $A \setminus B = \{1, 9\}$, $B \setminus A = \{2, 6, 8, 10, 12, 14\}$, so $A \oplus B = \{1, 2, 6, 8, 9, 10, 12, 14\}$.

- (b) $A \oplus B = (A \setminus B) \cup (B \setminus A)$ (definition of $\oplus$)
1. $= (A \cap \overline{B}) \cup (B \cap \overline{A})$ (set difference law)
 2. $= ((A \cap \overline{B}) \cup B) \cap ((A \cap \overline{B}) \cup \overline{A})$ (distributive law)
 3. $= ((A \cup B) \cap (\overline{B} \cup B)) \cap ((A \cup \overline{A}) \cap (\overline{B} \cup \overline{A}))$ (distributive, commutative laws)
 4. $= ((A \cup B) \cap U) \cap (U \cap (\overline{A} \cup \overline{B}))$ (commutative, complement laws)
 5. $= (A \cup B) \cap (\overline{A} \cup \overline{B})$ (commutative, identity laws)
 6. $= (A \cup B) \cap \overline{A \cap B}$ (De Morgan's law)
 7. $= (A \cup B) \setminus (A \cap B)$ (set difference law). ■

TRAP T3 Q3(c) · $A \times \mathcal{P}(B) \neq \mathcal{P}(A \times B)$

For every A, B , $\emptyset \in \mathcal{P}(A \times B)$ ($\emptyset \subseteq A \times B$), but $\emptyset \notin A \times \mathcal{P}(B)$ (every member of a Cartesian product is an ordered pair; $\emptyset$ is not one). This named witness proves the sets differ. A cardinality “proof” ($n \cdot 2^k \neq 2^{nk}$ for $|A| = n$, $|B| = k$) is wrong: equality holds when $n = 1$, or $n = 2$ and $k = 1$.

Relations and orders (Tutorials 4–5)**WORKED EXAMPLE T4 Q10 · Equivalence relation using Appendix A**

On $\mathbb{Z} \setminus \{0\}$ let $a \sim b \leftrightarrow ab > 0$. (a) Prove $\sim$ is an equivalence relation. (b) Determine all distinct equivalence classes.

1. (Reflexivity) Let $a \in \mathbb{Z} \setminus \{0\}$. Since $a \neq 0$, $a^2 > 0$ (T21). Thus $a \sim a$ (definition of $\sim$).
2. (Symmetry) Let $a, b \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$. Then $ab > 0$ (definition of $\sim$), so $ba > 0$ (commutative law of $\times$), so $b \sim a$.

3. (Transitivity) Let $a, b, c \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$ and $b \sim c$. Then $ab > 0$ and $bc > 0$ (definition of $\sim$).

3.1. Multiplying, $ab^2c > 0$ (Ord1), so $(ac)b^2 > 0$ (associative, commutative laws of $\times$).

3.2. So ac and b^2 are both positive or both negative (T25).

3.3. Since $b^2 > 0$ (T21, as $b \neq 0$), $ac > 0$. Thus $a \sim c$.

4. Therefore $\sim$ is an equivalence relation. ■

(b) T25 rules out pairs of opposite signs; the product of two positive or two negative integers is positive. The classes are $[1] = \mathbb{Z}^+$ and $[-1] = \mathbb{Z}^-$, so $(\mathbb{Z} \setminus \{0\})/\sim = \{\mathbb{Z}^+, \mathbb{Z}^-\}$.

T21: $a \neq 0 \Rightarrow a^2 > 0$. Ord1: $a, b > 0 \Rightarrow a + b > 0 \wedge ab > 0$. T25: $ab > 0 \Rightarrow a, b$ both positive or both negative.

WORKED EXAMPLE T4 Q9 · Inverse and composition by the element method

$S = \{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \text{ is even}\}$. Determine (a) S^{-1} , (b) $S \circ S$, (c) $S \circ S^{-1}$.

(a) $S^{-1} = \{(x, y) \in \mathbb{Z}^2 : (y, x) \in S\}$ (definition of inverse)

$= \{(x, y) \in \mathbb{Z}^2 : y^2 + x^2 \text{ is even}\}$ (definition of S)

$= \{(x, y) \in \mathbb{Z}^2 : x^2 + y^2 \text{ is even}\}$ (commutativity of $+$)

$= S$.

(b) $S \circ S = S$:

1. ($\subseteq$) Let $(x, z) \in S \circ S$. Then $(x, y) \in S$ and $(y, z) \in S$ for some $y \in \mathbb{Z}$ (definition of composition).

1.1. So $x^2 + y^2$ and $y^2 + z^2$ are even (definition of S), hence $x^2 + 2y^2 + z^2$ is even (sum of two even integers).

1.2. As $2y^2$ is even (definition of even, closure), $x^2 + z^2$ is even (difference of two even integers), so $(x, z) \in S$ (definition of S).

2. ($\supseteq$) Let $(x, z) \in S$. Note $(x, x) \in S$, as $x^2 + x^2$ is even. Since $(x, x) \in S$ and $(x, z) \in S$, $(x, z) \in S \circ S$ (definition of composition).

3. Therefore $S \circ S = S$. ■

(c) $S \circ S^{-1} = S \circ S = S$ by (a) and (b).

Alternative for ($\supseteq$): by cases. If x^2 is odd, so is z^2 , and $(x, 1), (1, z) \in S$. If x^2 is even, so is z^2 , and $(x, 0), (0, z) \in S$.

PAST PAPER T4 Q8 (AY20/21 exam) · T4 Q6 (AY23/24 exam)

Q8. On $\mathbb{Z}^+ \times \mathbb{Z}^+$, $(a, b) \sim (c, d) \leftrightarrow ab = cd$. Write the set-builder line first: $[(1, 1)]$

$= \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : (1, 1) \sim (x, y)\}$ (definition of class)

$= \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 1\}$ (definition of $\sim$)

$= \{(1, 1)\}$. Likewise $[(4, 3)] = \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 12\} = \{(1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1)\}$.

Q6. $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$ on $\{1, 2, 3\}$; find $R \circ R \circ \dots \circ R$ (7 copies). R is an equivalence relation, so $R \circ R = R$ (AY23 Q12, §23). By associativity, $R^7 = ((R \circ R) \circ (R \circ R)) \circ ((R \circ R) \circ R) = (R \circ R) \circ (R \circ R) = R \circ R = R$.

WORKED EXAMPLE T5 Q6 · Reflexive closure (the closure-proof template)

Let R be a relation on a non-empty set A , and define S on A by $xSy \leftrightarrow ((x = y) \vee (xRy))$. Show: (a) S is reflexive; (b) $R \subseteq S$; (c) if S' is a reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$. What is S called?

(a) Let $x \in A$. $x = x$, so xSx (definition of S). Therefore S is reflexive.

(b) Suppose $(x, y) \in R$. Then xSy (definition of S), so $(x, y) \in S$. Therefore $R \subseteq S$ (definition of $\subseteq$).

(c) Let $(x, y) \in S$: $x = y$ or xRy (definition of S).

1. **Case 1 ($x = y$):** $xS'y$ (S' is reflexive), so $(x, y) \in S'$.

2. **Case 2 (xRy):** $(x, y) \in R \subseteq S'$, so $(x, y) \in S'$.

In all cases $(x, y) \in S'$. Therefore $S \subseteq S'$. ■

S is the **reflexive closure** of R : the smallest reflexive relation on A containing R . Parts (a)–(c) mirror the three conditions defining a closure.

WORKED EXAMPLE T5 Q7 · Asymmetric vs antisymmetric

On $A = \{a, b, c\}$ find a relation that is (a) asymmetric and antisymmetric; (b) antisymmetric but not asymmetric; (c) asymmetric but not antisymmetric; (d) neither.

(a) $\{(a, b), (a, c)\}$. (b) $\{(a, a), (b, c)\}$. (c) **No solution:** every asymmetric relation is antisymmetric (below). (d) $\{(a, b), (b, a), (a, c)\}$.

1. Let R be an asymmetric relation on A : $\forall x, y \in A (xRy \rightarrow yRx)$ (definition of asymmetry).
 - 1.1. $\equiv \forall x, y \in A (xRy \vee yRx)$ (implication law).
 - 1.2. $\Rightarrow \forall x, y \in A ((xRy \vee yRx) \vee x = y)$ (generalization).
 - 1.3. $\equiv \forall x, y \in A (\sim(xRy \wedge yRx) \vee x = y)$ (De Morgan's law).
 - 1.4. $\equiv \forall x, y \in A (xRy \wedge yRx \rightarrow x = y)$ (implication law).
2. The final formula is the definition of antisymmetry, so R is antisymmetric: asymmetry makes it vacuously true. ■

TRAP T5 Q1 · Disprove with one named counterexample

On strings over $\{s, u\}$, $aRb \leftrightarrow \text{len}(a) \leq \text{len}(b)$ is **not** a partial order: sRu and uRs but $s \neq u$, so R is not antisymmetric. One counterexample suffices; there is no need to check reflexivity or transitivity.

Part VII · New & Updated Definition Index Entries (§32)

Absolute value: $|x| = x$ if $x \geq 0$, $-x$ if $x < 0$. §10

Argument; valid; sound, unsound: premises then conclusion; valid if true premises force a true conclusion; sound if valid with true premises, else unsound. §4

Colorful: $n = 3k$ for some $k \in \mathbb{Z}$ (lecture only). §10

Conditional: $p \rightarrow q$, false only for T $\rightarrow$ F; hypothesis p , conclusion q . §3

Critical row: truth-table row with all premises true. §4

Directed graph: one vertex per element; arrow $x \rightarrow y$ iff xRy . §19

Disjoint; mutually disjoint: $A \cap B = \emptyset$; $A_i \cap A_j = \emptyset$ for $i \neq j$. §16

Empty set; singleton: $\emptyset$, no elements; exactly one element. §14

Interval notation: (a, b) , $[a, b]$, $(a, b]$, $[a, b)$, $[a, \infty)$, ... as subsets of $\mathbb{R}$. §13

Lowest terms: a/b where 1 is the largest common divisor. §10

n -ary relation: subset of $A_1 \times \dots \times A_n$. §19

Negation, conjunction, disjunction: $\sim p$; $p \wedge q$; $p \vee q$. §1

Ordered pair; n -tuple: equal iff equal componentwise. §15

Partition; component: non-empty subsets; each element in exactly one; its elements. §18

Predicate; domain; truth set: sentence with variables; allowed values; $\{x \in D : P(x)\}$. §5

Rational / irrational: $r = a/b$ with $a, b \in \mathbb{Z}$, $b \neq 0$ / not rational. §10

Reflexive / symmetric closure: $R \cup \{(x, x) : x \in A\}$ / $R \cup R^{-1}$. §22

Relation; relation on A : subset of $A \times B$; subset of $A \times A$. §19

Representative: any member b of $[a]$; then $[b] = [a]$. §23

Set equality: $A \subseteq B$ and $B \subseteq A$. §14

Set-roster notation: elements listed in braces. §13

Subset; superset: $\forall x(x \in A \rightarrow x \in B)$; $B \supseteq A$. §14

Syllogism: two premises and a conclusion. §4

Uniqueness $\exists!$: exactly one element satisfies it. §5

Universal conditional; universal existential; existential universal: $\forall x(P \rightarrow Q)$; $\forall \exists$; $\exists \forall$. §5

Universal set: U , all objects under discussion. §16

Valid argument form: true premises force a true conclusion. §4, §8

Well-defined operation: binary operation on A inducing a rule on classes independent of representatives. §23

Well-ordered: total order in which every non-empty subset has a smallest element. §24